

4.1 Answer each of the following:

- a. Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2 .
- b. Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.
- c. Suppose that a simple regression has quantities $\sum (y_i - \bar{y})^2 = 631.63$ and $\sum \hat{e}_i^2 = 182.85$, find R^2 .

a.

$$R^2 = \frac{SSR}{SST} = \frac{\sum (\hat{y}_i - \bar{y})^2}{\sum (y_i - \bar{y})^2} = \frac{375.47}{\sum y_i^2 - 2 \bar{y}^2 * 20 + \sum \bar{y}^2}$$

$$= \frac{375.47}{7825.94 - 2 * 19.21^2 * 20 + 20 * 19.21^2} = 0.843$$

b.

$$\hat{\sigma}^2 = \sum \frac{(y_i - b_1 - b_2 x_i)^2}{N - 2} = \frac{\sum (y_i - \hat{y}_i)^2}{N - 2} = \frac{SSE}{18}$$

$$R^2 = 0.7911 = 1 - \frac{SSE}{SST} = 1 - \frac{SSE}{725.94} \Rightarrow SSE = 151.6489$$

$$\Rightarrow \hat{\sigma}^2 = \frac{151.6489}{18} = 8.42$$

c. $R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{182.85}{631.63} = 0.711$

4.3 We have five observations on x and y . They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^5 \hat{e}_i^2 = 3.6$, $\sum_{i=1}^5 (x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^5 (y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

- a. the predicted value of y for $x_0 = 4$.
- b. the $se(f)$ corresponding to part (a).
- c. a 95% prediction interval for y given $x_0 = 4$.
- d. a 99% prediction interval for y given $x_0 = 4$.
- e. a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

a. $1.2 + 0.8 * 4 = 4.4$

b. $\bar{x} = 1$

$$\text{var}(f) = \hat{\sigma}^2 \left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right] = \frac{3.6}{5 - 2} \left[1 + \frac{1}{5} + \frac{(4 - 1)^2}{10} \right] = 1.2 * 2.1$$

$$= 2.52$$

$$se(f) = \sqrt{2.52} = 1.59$$

c. $t_{0.975,3} = 3.18$

$$PI(95\%) = [4.4 - 3.18 * 1.59, 4.4 + 3.18 * 1.59] = [-0.6562, 9.4562]$$

d. $t_{0.995,3} = 5.84$

$$PI(99\%) = [4.4 - 5.84 * 1.59, 4.4 + 5.84 * 1.59] = [-4.8856, 13.6856]$$

e. $\hat{y} = 1.2 + 0.8 * 1 = 2$

$$\text{var}(f) = \frac{3.6}{5 - 2} \left[1 + \frac{1}{5} \right] = 1.2 * 1.2 = 1.44$$

$$se(f) = 1.2$$

$$PI(95\%, \bar{x}) = [2 - 3.18 * 1.2, 2 + 3.18 * 1.2] = [-1.816, 5.816] \text{ 寬度更小了。}$$

4.25 The file *collegetown* contains data on 500 houses sold in Baton Rouge, LA during 2009–2013. Variable descriptions are in the file *collegetown.def*.

- a. Estimate the log-linear model $\ln(PRICE) = \beta_1 + \beta_2 SQFT + e$. Interpret the estimated model parameters. Calculate the slope and elasticity at the sample means, if necessary.
- b. Estimate the log-log model $\ln(PRICE) = \alpha_1 + \alpha_2 \ln(SQFT) + e$. Interpret the estimated parameters. Calculate the slope and elasticity at the sample means, if necessary.
- c. Compare the R^2 value from the linear model $PRICE = \delta_1 + \delta_2 SQFT + e$ to the “generalized” R^2 measure for the models in (b) and (c).
- d. Construct histograms of the least squares residuals from each of the models in (a)–(c) and obtain the Jarque–Bera statistics. Based on your observations, do you consider the distributions of the residuals to be compatible with an assumption of normality?
- e. For each of the models in (a)–(c), plot the least squares residuals against $SQFT$. Do you observe any patterns?
- f. For each model in (a)–(c), predict the value of a house with 2700 square feet.
- g. For each model in (a)–(c), construct a 95% prediction interval for the value of a house with 2700 square feet.
- h. Based on your work in this problem, discuss the choice of functional form. Which functional form would you use? Explain.

a. $b1 = 4.39$: the average of $\ln(\text{price})=4.39$ when $\text{sqft}=0$

$b2 = 0.036$: sqft increase 1 unit, price increase 3.6%

slope = $\beta_2 * \text{price} = 0.036 * 250.24 = 9.01$ (average of $\text{price}=250.24$)

elasticity = $\frac{dy/y}{dx/x} = \beta_2 * \text{sqft} = 0.036 * 27.28 = 0.982$ (average of $\text{sqft} = 27.28$)

b. $a1 = 2.05$: the average of $\ln(\text{price})=2.05$ when $\ln(\text{sqft})=0$

$a2 = 1.025$: sqft increase 1%, price increase 1.025%

slope = $\frac{dy}{dx} = \alpha_2 \frac{\text{price}}{\text{sqft}} = 1.025 * \frac{250.24}{27.28} = 9.4$

elasticity = $\frac{dy/y}{dx/x} = \alpha_2 = 1.025$

c. $d1=-115.42, d2=13.4$

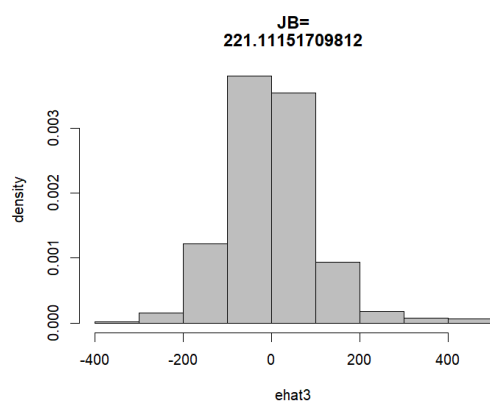
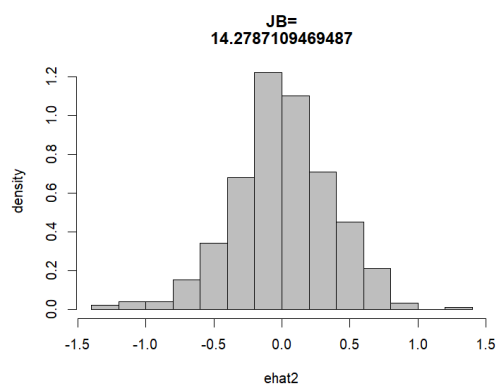
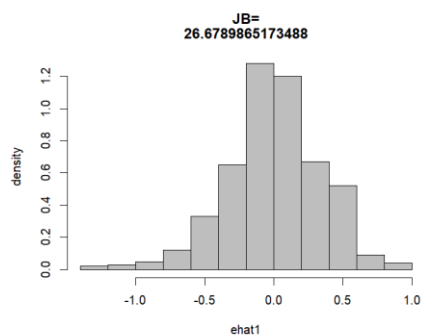
$R^2 = 0.641$

$R^2_{g,(c)} = 0.641$

$R^2_{g,(b)} = 0.645$

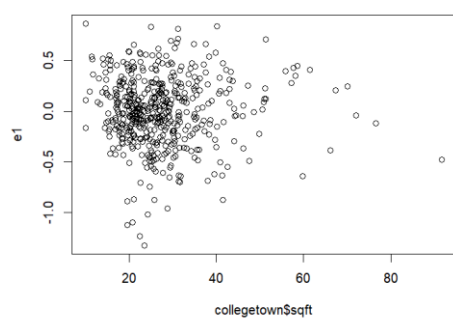
d.

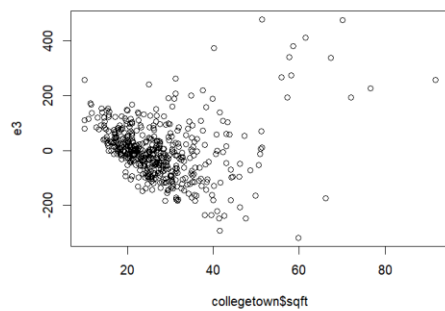
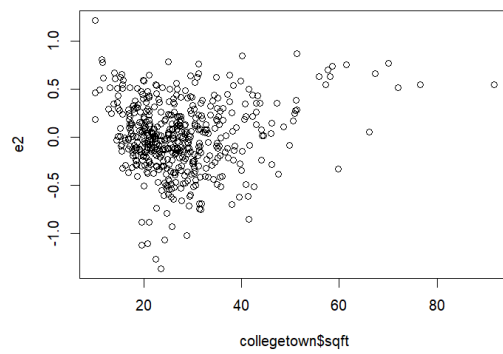
三個模型的 residual 都不是 normal，因為沒人通過 JB test，但只有模型 2 的 hist 有點像是常態分佈。



e.

Residuals 集中在 $\text{sqft} < 40$ 的部分(a)和(b)的左半部分 var 大，右半 var 小。
但是(c)當 $\text{sqft} > 40$ 以後平均不是 0。





- f. $\text{Predict(a)} = \exp(4.39 + 0.036 * 27) = 213.15$
 $\text{Predict(b)} = \exp(2.05 + 1.025 * \log(27)) = 227.75$
 $\text{Predict(c)} = -115.42 + 13.4 * 27 = 246.38$
- g. $\text{CI(a): } [109.7685, 418.1167]$
 $\text{CI(b): } [111.1406, 465.8406]$
 $\text{CI(c): } [44.27727, 448.6341]$
- h. 我會使用 `model(b)`，因為他的 `residual` 經過 `JBtest` 最接近常態，畫出來的 `residual` 也很像是常態分配。